

FINAL EXAMINATION

DevOps, Cloud Computing, Kubernetes, Terraform & Web Security

Multiple Choice Questions (MCQ)

Instructions: This exam contains 110 multiple-choice questions. Select the BEST answer for each question. Each question is worth 1 point. Total marks: 110. Good luck!

Student Name: _____ ID: _____ Date: _____

Section: DevOps & CI/CD

1. What does DevOps combine?

- A) Development and Operations
- B) Design and Optimization
- C) Debugging and Outsourcing
- D) Deployment and Observation

2. Which of the following is NOT a core DevOps principle?

- A) Automation of the SDLC
- B) Collaboration and communication
- C) Waterfall project management
- D) Continuous improvement and minimization of waste

3. What does CI stand for in CI/CD?

- A) Continuous Inspection
- B) Continuous Integration
- C) Code Integration
- D) Centralized Infrastructure

4. What is the main difference between Continuous Delivery and Continuous Deployment?

- A) Continuous Delivery is faster
- B) Continuous Deployment requires manual testing
- C) Continuous Delivery requires a manual approval to deploy to production
- D) There is no difference

5. What does 'Shift Left' mean in DevOps?

- A) Move deployment earlier
- B) Shift security and testing earlier in the development process
- C) Move the development team to the left side of the office
- D) Shift from cloud to on-premises

6. How does DevSecOps relate to DevOps?

- A) DevSecOps replaces DevOps entirely
- B) DevSecOps embeds security checks into every stage of the development lifecycle
- C) DevSecOps removes the need for testing
- D) DevSecOps only focuses on deployment security

7. Before CI/CD, what was a common problem during code merging?

- A) Automated conflict resolution
- B) Big conflicts and broken builds from merging long-lived branches
- C) Instant deployment
- D) No need for testing

8. What is a CI/CD Pipeline?

- A) A methodology for writing code
- B) A series of automatic steps that build, test, and deliver code

- C) A type of version control system
- D) A manual deployment checklist

9. Which deployment strategy involves running the old and new versions simultaneously?

- A) Waterfall deployment
- B) Big bang deployment
- C) Blue-green deployment
- D) Manual deployment

10. In a CI/CD pipeline, what happens in the 'Verify' phase?

- A) Code is deployed to production
- B) Code is tested to ensure correctness and quality standards
- C) Users are notified
- D) Servers are configured

Section: Containers & Docker

11. What is the main purpose of Docker?

- A) To virtualize different operating systems
- B) To containerize applications and ship them
- C) To replace virtual machines entirely
- D) To manage network infrastructure

12. What do Docker containers share with the host?

- A) The entire operating system
- B) The OS kernel
- C) The application code
- D) The user interface

13. Can you run a Windows-based container on a Linux Docker host?

- A) Yes, always
- B) Yes, with special plugins
- C) No, because they don't share the same kernel
- D) Yes, but only in production

14. How do containers differ from Virtual Machines in terms of size?

- A) Containers are typically in Gigabytes
- B) Containers are typically in Megabytes, VMs in Gigabytes
- C) They are the same size
- D) VMs are smaller than containers

15. What is a Docker image?

- A) A running container
- B) A package or template used to create containers
- C) A virtual machine snapshot

- D) A network configuration file

16. Which Dockerfile instruction specifies the base image?

- A) RUN
- B) CMD
- C) FROM
- D) COPY

17. What does the WORKDIR instruction do in a Dockerfile?

- A) Copies files to the container
- B) Sets the working directory inside the container
- C) Exposes a port
- D) Runs a command

18. What is the difference between CMD and RUN in a Dockerfile?

- A) CMD runs during build, RUN runs at startup
- B) RUN executes during build, CMD specifies the default command at container startup
- C) They are identical
- D) CMD installs packages, RUN starts services

19. What is Container Orchestration?

- A) Writing Dockerfiles
- B) Automatically deploying and managing containers at scale
- C) Creating Docker images
- D) Installing Docker on servers

20. Which of the following is a container orchestration tool?

- A) Git
- B) Jenkins
- C) Kubernetes
- D) Ansible

Section: Kubernetes

21. What is the smallest deployable unit in Kubernetes?

- A) Container
- B) Node
- C) Pod
- D) Deployment

22. What component in K8s architecture acts as the front-end for the cluster?

- A) ETCD
- B) Kubelet
- C) API Server
- D) Scheduler

23. What is ETCD used for in Kubernetes?

- A) Container runtime
- B) Distributed reliable key-value store for cluster data
- C) Load balancing
- D) DNS resolution

24. What does the Kubernetes Scheduler do?

- A) Monitors node health
- B) Distributes containers across multiple nodes
- C) Manages networking
- D) Stores cluster state

25. What is the role of kubelet?

- A) It is the cluster's database
- B) It runs on each node and ensures containers are running as expected
- C) It balances network traffic
- D) It schedules pods

26. Which Kubernetes Service type exposes a service only within the cluster?

- A) NodePort
- B) LoadBalancer
- C) ClusterIP
- D) ExternalName

27. What is the valid port range for a NodePort service in Kubernetes?

- A) 1-1024
- B) 8000-9000
- C) 30000-32767
- D) 80-443

28. What does Kubernetes Ingress manage?

- A) Internal pod-to-pod communication
- B) External HTTP/HTTPS access to services in the cluster
- C) Container image storage
- D) Node scaling

29. Why is putting WordPress and MySQL in the same Pod considered an anti-pattern?

- A) They can't communicate over a network
- B) They need different scaling strategies and can work on different machines
- C) WordPress requires Windows
- D) MySQL doesn't run in containers

30. What happens when a Liveness Probe fails repeatedly?

- A) The Pod is deleted
- B) The container is killed and restarted within the Pod
- C) Traffic is stopped
- D) The node is drained

31. What happens when a Readiness Probe fails?

- A) The container is restarted
- B) The Pod is removed from Service load balancing but keeps running
- C) The node shuts down
- D) The Pod is deleted

32. In Kubernetes resource management, what does 'requests' define?

- A) Maximum allowed CPU usage
- B) Minimum guaranteed resources used for scheduling
- C) Network bandwidth limits
- D) Storage capacity

33. What happens when a container exceeds its memory limit?

- A) It is throttled
- B) It is killed (OOMKilled)
- C) Nothing happens
- D) It gets more memory automatically

34. What is the purpose of a Kubernetes ConfigMap?

- A) To store encrypted secrets
- B) To store non-confidential configuration data as key-value pairs
- C) To define resource limits
- D) To manage container images

35. How should data in a Kubernetes Secret be encoded?

- A) Plain text
- B) Base64 encoded
- C) SHA-256 hashed
- D) AES encrypted

36. What does a ReplicaSet ensure?

- A) Each pod has unique configuration
- B) A stable set of a desired number of pods is always running
- C) Pods are deployed to specific nodes
- D) Network policies are enforced

37. What does a Kubernetes Deployment manage that a ReplicaSet doesn't?

- A) Pod creation
- B) Rolling updates, rollbacks, and revision history
- C) Network routing
- D) Storage volumes

38. Which kubectl command applies a declarative configuration file?

- A) kubectl run
- B) kubectl create
- C) kubectl apply -f
- D) kubectl exec

Section: Terraform & IaC

39. What is Infrastructure as Code (IaC)?

- A) Manually configuring servers
- B) Provisioning and managing infrastructure using code instead of manual configuration
- C) Writing application code
- D) Documenting infrastructure in spreadsheets

40. What does 'idempotent' mean in the context of IaC?

- A) Running the script once produces errors
- B) Running the script multiple times always results in the same desired state
- C) The script can only run once
- D) The script modifies state randomly

41. What is the difference between mutable and immutable infrastructure?

- A) There is no difference
- B) Mutable infrastructure is updated in place; immutable replaces servers with new ones
- C) Immutable infrastructure is always updated in place
- D) Mutable infrastructure cannot be changed

42. Terraform uses which approach to define infrastructure?

- A) Imperative
- B) Procedural
- C) Declarative
- D) Object-oriented

43. What language does Terraform use for configuration?

- A) YAML
- B) JSON only
- C) HCL (HashiCorp Configuration Language)
- D) Python

44. What does 'terraform init' do?

- A) Deploys infrastructure
- B) Destroys resources
- C) Scans code, downloads provider plugins, and initializes the working directory
- D) Shows the execution plan

45. What does 'terraform plan' do?

- A) Creates resources immediately
- B) Shows what changes Terraform will make without applying them
- C) Destroys all resources
- D) Downloads provider plugins

46. In Terraform plan output, what does a plus sign (+) indicate?

- A) A resource will be deleted
- B) A resource will be modified
- C) A resource will be created
- D) A resource is unchanged

47. What is the correct syntax for a Terraform resource block?

- A) resource "PROVIDER_TYPE" "NAME" { CONFIG }
- B) create "TYPE" "NAME" { CONFIG }
- C) deploy "PROVIDER" "TYPE" { CONFIG }
- D) instance "NAME" { CONFIG }

48. What is a Terraform data source?

- A) A way to create new resources
- B) Read-only information fetched from the provider
- C) A database connection
- D) A log file

49. How do you reference a variable in Terraform?

- A) \${variable_name}
- B) var.variable_name
- C) @variable_name
- D) \$variable_name

50. What is the purpose of Terraform output variables?

- A) To define input parameters
- B) To export useful information (like IP addresses) after apply
- C) To create resources
- D) To delete state files

51. What is the default Terraform state file name?

- A) state.json
- B) terraform.tfstate
- C) main.tf
- D) terraform.lock.hcl

52. Why is storing Terraform state in version control a bad idea?

- A) It's too large
- B) Manual errors, no locking, and secrets stored in plain text
- C) Git doesn't support JSON
- D) It's automatically backed up anyway

53. What is a recommended remote backend for Terraform state?

- A) Local filesystem
- B) Git repository
- C) Amazon S3 with DynamoDB for locking
- D) Email attachment

54. What is the primary key name required for a DynamoDB table used for Terraform state locking?

- A) StateID
- B) LockID
- C) TerraformLock
- D) lock_key

55. What is a Terraform module?

- A) A single resource block
- B) A reusable, self-contained package of Terraform configuration
- C) A provider plugin
- D) A state file

56. What is the typical file structure of a Terraform module?

- A) main.tf, variables.tf, outputs.tf
- B) index.js, package.json
- C) Dockerfile, docker-compose.yml
- D) app.py, requirements.txt

57. What does the lifecycle setting 'create_before_destroy' do in Terraform?

- A) Deletes the resource first, then creates a new one
- B) Creates the replacement resource before deleting the old one
- C) Prevents any changes to the resource
- D) Skips the resource during apply

Section: Web Application Security

58. What is the fundamental security problem with web applications?

- A) Slow performance
- B) All user input is untrusted
- C) Too many features
- D) Expensive hosting

59. What is SQL Injection?

- A) A method to optimize SQL queries
- B) An attack where malicious SQL is inserted into input fields to manipulate database queries
- C) A database backup technique
- D) A type of database encryption

60. In SQL injection, what does the '--' sequence do?

- A) Starts a new query
- B) Acts as a comment, removing the rest of the query
- C) Encrypts the query
- D) Creates a new table

61. What is the purpose of the UNION keyword in SQL injection attacks?

- A) To delete data
- B) To combine results from multiple SELECT queries to extract additional data
- C) To update records
- D) To create tables

62. Why is NULL used when determining column count in UNION-based SQL injection?

- A) NULL is faster to process
- B) NULL is type-flexible and can fit into almost any column type
- C) NULL deletes data
- D) NULL is required by SQL syntax

63. What is Out-of-Band SQL Injection?

- A) SQL injection via the URL
- B) Exploiting SQL injection by triggering network interactions (like DNS) to exfiltrate data
- C) SQL injection that crashes the server
- D) SQL injection in stored procedures only

64. What is OS Command Injection?

- A) Installing malware via email
- B) Injecting shell commands through application inputs that are passed to system commands
- C) Brute forcing SSH passwords
- D) Modifying system configuration files directly

65. In command injection, what does the '&' character do?

- A) Combines two strings
- B) Acts as a shell command separator to execute multiple commands
- C) Encrypts the command
- D) Redirects output to a file

66. What is a Path Traversal attack?

- A) Navigating a website normally
- B) Using '../' sequences to access files outside the intended directory
- C) A type of network scanning
- D) A DoS attack

67. What is SSRF (Server-Side Request Forgery)?

- A) A client-side scripting attack
- B) An attack that causes the server to make requests to unintended locations
- C) A database encryption method
- D) A password cracking technique

68. In SSRF, why do applications trust requests from localhost (127.0.0.1)?

- A) Localhost is always secure
- B) Access controls may be bypassed for requests from the local machine
- C) Localhost has special encryption
- D) Localhost uses a different protocol

69. What is Cross-Site Scripting (XSS)?

- A) Injecting SQL into a database
- B) Injecting malicious scripts into trusted websites that execute in victims' browsers
- C) A network protocol attack
- D) A server-side vulnerability only

70. What are the three interrelated mechanisms for handling user access in web applications?

- A) Firewall, VPN, Encryption
- B) Authentication, Session Management, Access Control
- C) Login, Logout, Password Reset
- D) HTTP, HTTPS, SSH

71. What is a session token?

- A) A permanent user identifier
- B) A unique string that maps to a user's session on the server
- C) A type of encryption key
- D) A database record

72. What is 'salting' in password storage?

- A) Encrypting the entire database
- B) Adding a unique random string to each password before hashing
- C) Using a stronger hash algorithm
- D) Storing passwords in plain text with a label

73. What is a Key Vault used for?

- A) Physical server security
- B) Managing secrets, encryption keys, passwords, and certificates with controlled access
- C) Load balancing
- D) DNS management

74. What does TLS provide?

- A) Only encryption
- B) Data encryption, data integrity, and authentication
- C) Only authentication
- D) Only data integrity

75. What is the purpose of a digital certificate?

- A) To encrypt data at rest
- B) To link a public key to an entity, verified by a trusted certificate authority
- C) To store passwords
- D) To configure firewalls

76. Which of the following is NOT a recommended file upload mitigation?

- A) Check file extension
- B) Check MIME type against a whitelist
- C) Allow any file type and size
- D) Use antivirus scanning

Section: AWS Cloud

77. What is cloud computing?

- A) Hosting applications on your local machine
- B) On-demand delivery of IT resources over the internet with pay-as-you-go pricing
- C) Buying physical servers
- D) Using only open-source software

78. Which cloud model uses dedicated resources for a single organization?

- A) Public cloud
- B) Hybrid cloud
- C) Private cloud
- D) Community cloud

79. What is a hybrid cloud?

- A) A cloud only for startups
- B) A combination of public and private clouds with data/applications moving between them
- C) A cloud with no internet access
- D) A cloud that uses only Linux servers

80. What does SaaS stand for?

- A) Storage as a Service
- B) Software as a Service
- C) Security as a Service
- D) Server as a Service

81. In which cloud service model does the developer only focus on writing and deploying code?

- A) IaaS
- B) SaaS
- C) PaaS
- D) FaaS

82. What is an AWS Region?

- A) A single data center
- B) A separate geographic area containing multiple Availability Zones
- C) A type of server
- D) A pricing model

83. How many Availability Zones does each AWS Region have at minimum?

- A) 1
- B) 2
- C) 3
- D) 5

84. What is Amazon EC2?

- A) A storage service

- B) A virtual machine in the cloud
- C) A database service
- D) A DNS service

85. What is the hypervisor's role in EC2?

- A) Storing data
- B) Managing resource sharing and isolation between virtual machines
- C) Routing network traffic
- D) Encrypting data

86. What type of EC2 instances are ideal for compute-intensive tasks?

- A) General purpose
- B) Memory optimized
- C) Compute optimized
- D) Storage optimized

87. What is AWS Lambda?

- A) A virtual machine service
- B) A serverless compute service that runs code without managing servers
- C) A storage service
- D) A database service

88. What is a Lambda 'cold start'?

- A) When Lambda runs out of memory
- B) Latency from creating a new execution environment for a Lambda function
- C) When Lambda crashes
- D) When Lambda is deleted

89. What is the default concurrency limit for Lambda functions per Region?

- A) 100
- B) 500
- C) 1000
- D) 5000

90. What is Amazon S3?

- A) A relational database
- B) An object storage service
- C) A compute service
- D) A networking service

91. What is the maximum size of a single object in S3?

- A) 1 GB
- B) 5 GB
- C) 5 TB
- D) Unlimited

92. Which S3 storage class is the lowest cost for archival?

- A) S3 Standard

- B) S3 Intelligent-Tiering
- C) S3 Glacier Deep Archive
- D) S3 One Zone-IA

93. What is Amazon VPC?

- A) A compute service
- B) A logically isolated virtual network in AWS
- C) A database service
- D) A monitoring tool

94. What makes a subnet 'public' in AWS?

- A) Having EC2 instances in it
- B) Its route table having a route to an Internet Gateway
- C) Having a public IP range
- D) Being in a specific Availability Zone

95. What is the purpose of a NAT Gateway?

- A) Allow inbound internet traffic to private subnets
- B) Allow private subnet instances to access the internet without being reachable from it
- C) Load balance traffic
- D) Encrypt data in transit

96. What is the difference between Security Groups and Network ACLs?

- A) Security Groups are stateless, NACLs are stateful
- B) Security Groups are stateful and instance-level; NACLs are stateless and subnet-level
- C) They are identical
- D) NACLs are applied to instances

97. What is an Application Load Balancer (ALB)?

- A) A Layer 4 load balancer
- B) A Layer 7 load balancer that routes based on HTTP content (URLs, headers, hostnames)
- C) A DNS service
- D) A database proxy

98. What is cross-zone load balancing?

- A) Load balancing between AWS Regions
- B) Each LB node distributes traffic across targets in all enabled Availability Zones
- C) Load balancing between on-premises and cloud
- D) Distributing traffic by IP address only

99. What does an Auto Scaling Group (ASG) do?

- A) Distributes traffic across instances
- B) Automatically launches or terminates EC2 instances to maintain desired capacity
- C) Manages DNS records
- D) Encrypts data at rest

100. In an ASG, what is the 'desired capacity'?

- A) The maximum number of instances

- B) The target number of instances the ASG tries to maintain
- C) The minimum number of instances
- D) The number of failed instances

101. What is Amazon RDS?

- A) A NoSQL database service
- B) A managed relational database service
- C) An object storage service
- D) A container orchestration service

102. What is the benefit of RDS Multi-AZ deployment?

- A) Lower cost
- B) Automatic failover and data redundancy across Availability Zones
- C) Faster query performance
- D) More storage space

103. What is VPC Peering?

- A) A transitive networking connection
- B) A direct, non-transitive networking connection between two VPCs using private IPs
- C) A public internet connection
- D) A type of load balancer

104. What is an AWS Transit Gateway?

- A) A type of EC2 instance
- B) A central hub that connects multiple VPCs and on-premises networks
- C) A storage gateway
- D) A serverless compute service

105. What does IAM stand for in AWS?

- A) Internet Access Management
- B) Identity and Access Management
- C) Infrastructure as a Module
- D) Integrated Application Monitor

106. Which EC2 pricing model offers the largest discount but can be interrupted?

- A) On-Demand
- B) Reserved Instances
- C) Spot Instances
- D) Dedicated Hosts

107. What is the difference between synchronous and asynchronous replication in RDS?

- A) They are the same
- B) Synchronous waits for replica confirmation before acknowledging; asynchronous responds immediately
- C) Asynchronous is always faster
- D) Synchronous is only for read replicas

108. What is EBS (Elastic Block Store)?

- A) Object storage
- B) Block-level storage volumes attached to EC2 instances
- C) File storage shared across instances
- D) Archive storage

109. What is the difference between a Network Load Balancer and an Application Load Balancer?

- A) NLB is Layer 7, ALB is Layer 4
- B) NLB operates at Layer 4 (TCP/UDP) without inspecting HTTP; ALB operates at Layer 7 (HTTP)
- C) They are identical
- D) ALB is for non-HTTP traffic

110. In a default VPC, what CIDR block is used?

- A) 10.0.0.0/16
- B) 192.168.0.0/16
- C) 172.31.0.0/16
- D) 172.16.0.0/12

ANSWER KEY

DevOps & CI/CD

1. **A** - Development and Operations
2. **C** - Waterfall project management
3. **B** - Continuous Integration
4. **C** - Continuous Delivery requires a manual approval to deploy to production
5. **B** - Shift security and testing earlier in the development process
6. **B** - DevSecOps embeds security checks into every stage of the development lifecycle
7. **B** - Big conflicts and broken builds from merging long-lived branches
8. **B** - A series of automatic steps that build, test, and deliver code
9. **C** - Blue-green deployment
10. **B** - Code is tested to ensure correctness and quality standards

Containers & Docker

11. **B** - To containerize applications and ship them
12. **B** - The OS kernel
13. **C** - No, because they don't share the same kernel
14. **B** - Containers are typically in Megabytes, VMs in Gigabytes
15. **B** - A package or template used to create containers
16. **C** - FROM
17. **B** - Sets the working directory inside the container
18. **B** - RUN executes during build, CMD specifies the default command at container startup
19. **B** - Automatically deploying and managing containers at scale
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Kubernetes

21. **C** - Pod
22. **C** - API Server
23. **B** - Distributed reliable key-value store for cluster data
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32. **B** - Minimum guaranteed resources used for scheduling
33. **B** - It is killed (OOMKilled)
34. **B** - To store non-confidential configuration data as key-value pairs
35. **B** - Base64 encoded
36. **B** - A stable set of a desired number of pods is always running
37. **B** - Rolling updates, rollbacks, and revision history
38. **C** - kubectl apply -f

Terraform & IaC

39. **B** - Provisioning and managing infrastructure using code instead of manual configuration
40. **B** - Running the script multiple times always results in the same desired state
41. **B** - Mutable infrastructure is updated in place; immutable replaces servers with new ones
42. **C** - Declarative
43. **C** - HCL (HashiCorp Configuration Language)

- 44. **C** - Scans code, downloads provider plugins, and initializes the working directory
- 45. **B** - Shows what changes Terraform will make without applying them
- 46. **C** - A resource will be created
- 47. **A** - resource "PROVIDER_TYPE" "NAME" { CONFIG }
- 48. **B** - Read-only information fetched from the provider
- 49. **B** - var.variable_name
- 50. **B** - To export useful information (like IP addresses) after apply
- 51. **B** - terraform.tfstate
- 52. **B** - Manual errors, no locking, and secrets stored in plain text
- 53. **C** - Amazon S3 with DynamoDB for locking
- 54. **B** - LockID
- 55. **B** - A reusable, self-contained package of Terraform configuration
- 56. **A** - main.tf, variables.tf, outputs.tf
- 57. **B** - Creates the replacement resource before deleting the old one

Web Application Security

- 58. **B** - All user input is untrusted
- 59. **B** - An attack where malicious SQL is inserted into input fields to manipulate database queries
- 60. **B** - Acts as a comment, removing the rest of the query
- 61. **B** - To combine results from multiple SELECT queries to extract additional data
- 62. **B** - NULL is type-flexible and can fit into almost any column type
- 63. **B** - Exploiting SQL injection by triggering network interactions (like DNS) to exfiltrate data
- 64. **B** - Injecting shell commands through application inputs that are passed to system commands
- 65. **B** - Acts as a shell command separator to execute multiple commands
- 66. **B** - Using '../' sequences to access files outside the intended directory
- 67. **B** - An attack that causes the server to make requests to unintended locations
- 68. **B** - Access controls may be bypassed for requests from the local machine
- 69. **B** - Injecting malicious scripts into trusted websites that execute in victims' browsers
- 70. **B** - Authentication, Session Management, Access Control
- 71. **B** - A unique string that maps to a user's session on the server
- 72. **B** - Adding a unique random string to each password before hashing
- 73. **B** - Managing secrets, encryption keys, passwords, and certificates with controlled access
- 74. **B** - Data encryption, data integrity, and authentication
- 75. **B** - To link a public key to an entity, verified by a trusted certificate authority
- 76. **C** - Allow any file type and size

AWS Cloud

- 77. **B** - On-demand delivery of IT resources over the internet with pay-as-you-go pricing
- 78. **C** - Private cloud
- 79. **B** - A combination of public and private clouds with data/applications moving between them
- 80. **B** - Software as a Service
- 81. **C** - PaaS
- 82. **B** - A separate geographic area containing multiple Availability Zones
- 83. **C** - 3
- 84. **B** - A virtual machine in the cloud
- 85. **B** - Managing resource sharing and isolation between virtual machines
- 86. **C** - Compute optimized
- 87. **B** - A serverless compute service that runs code without managing servers
- 88. **B** - Latency from creating a new execution environment for a Lambda function
- 89. **C** - 1000
- 90. **B** - An object storage service
- 91. **C** - 5 TB
- 92. **C** - S3 Glacier Deep Archive
- 93. **B** - A logically isolated virtual network in AWS

- 94. **B** - *Its route table having a route to an Internet Gateway*
- 95. **B** - *Allow private subnet instances to access the internet without being reachable from it*
- 96. **B** - *Security Groups are stateful and instance-level; NACLs are stateless and subnet-level*
- 97. **B** - *A Layer 7 load balancer that routes based on HTTP content (URLs, headers, hostnames)*
- 98. **B** - *Each LB node distributes traffic across targets in all enabled Availability Zones*
- 99. **B** - *Automatically launches or terminates EC2 instances to maintain desired capacity*
- 100. **B** - *The target number of instances the ASG tries to maintain*
- 101. **B** - *A managed relational database service*
- 102. **B** - *Automatic failover and data redundancy across Availability Zones*
- 103. **B** - *A direct, non-transitive networking connection between two VPCs using private IPs*
- 104. **B** - *A central hub that connects multiple VPCs and on-premises networks*
- 105. **B** - *Identity and Access Management*
- 106. **C** - *Spot Instances*
- 107. **B** - *Synchronous waits for replica confirmation before acknowledging; asynchronous responds immediately*
- 108. **B** - *Block-level storage volumes attached to EC2 instances*
- 109. **B** - *NLB operates at Layer 4 (TCP/UDP) without inspecting HTTP; ALB operates at Layer 7 (HTTP)*
- 110. **C** - *172.31.0.0/16*